Ethanol Blending in Gasoline – Belgium

September 2025

U.S. GRAINS & BIOPRODUCTS COUNCIL

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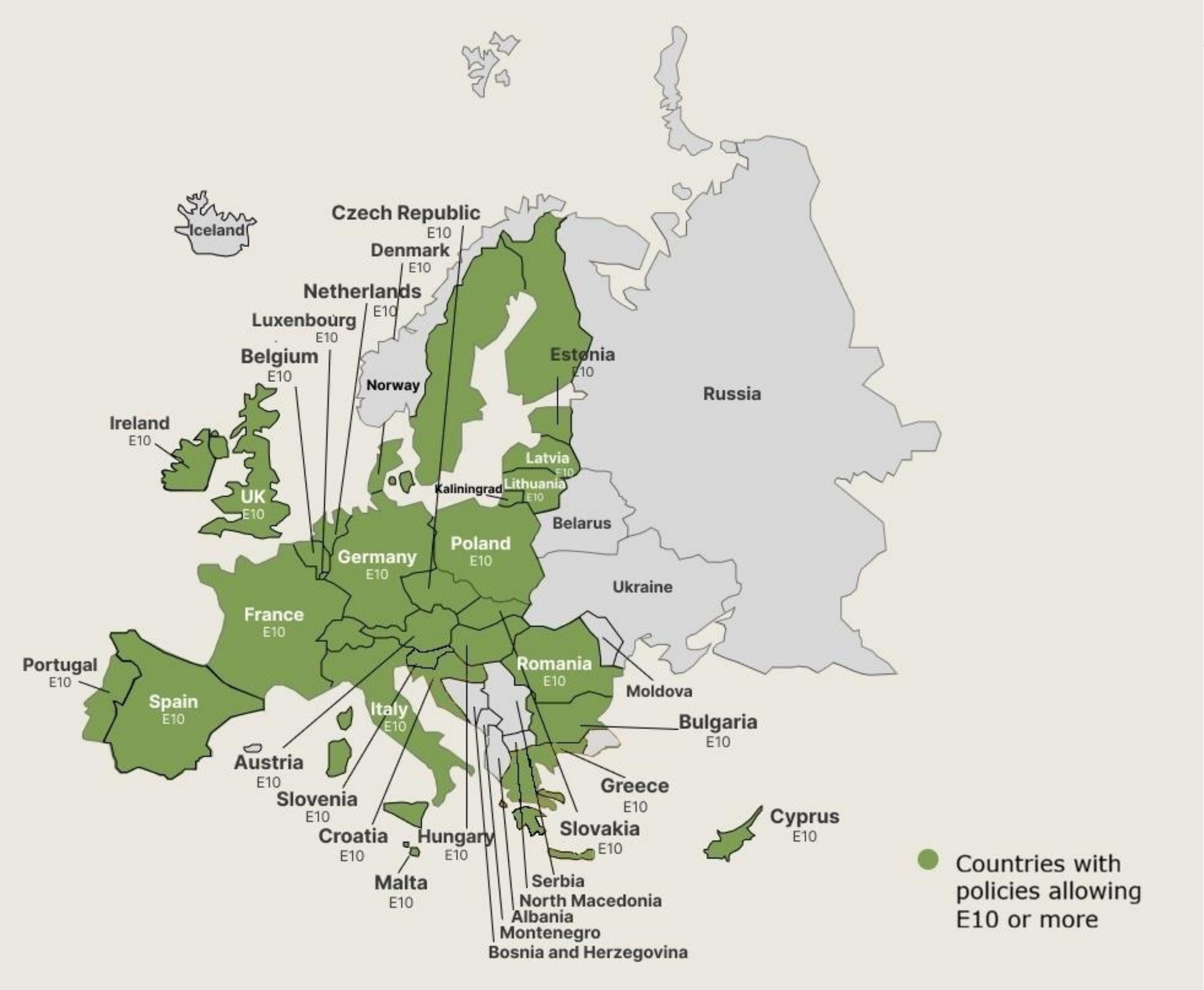
Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

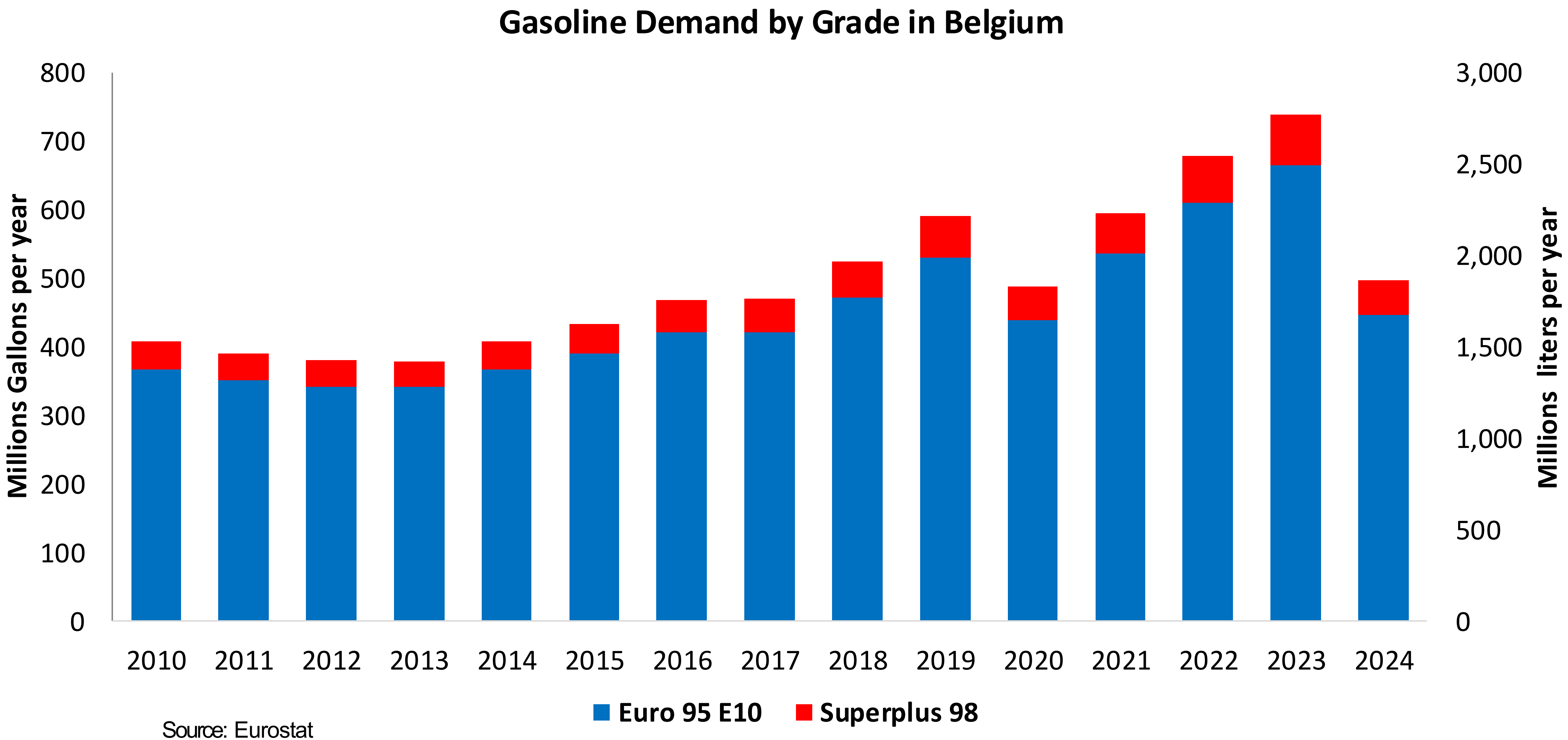
Ethanol Gasoline Blending in Belgium

In 2024, gasoline consumption in Belgium was 1,865 million liters (493 million gallons) and growth rate was -3.4%. Gasoline production and net imports were 5,257 and -4,825 million liters (1,389 and -1,275 million gallons) correspondingly. Belgium complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.3 RON.

In 2024, ethanol consumption in Belgium was 239 million liters (63 million gallons) representing 12.8% of gasoline demand. Ethanol production and Net Imports were 554 and -315 million liters (63 and -83 million gallons) correspondingly. Ethanol sources are Sugar Cane 20% and Cereals 80%.

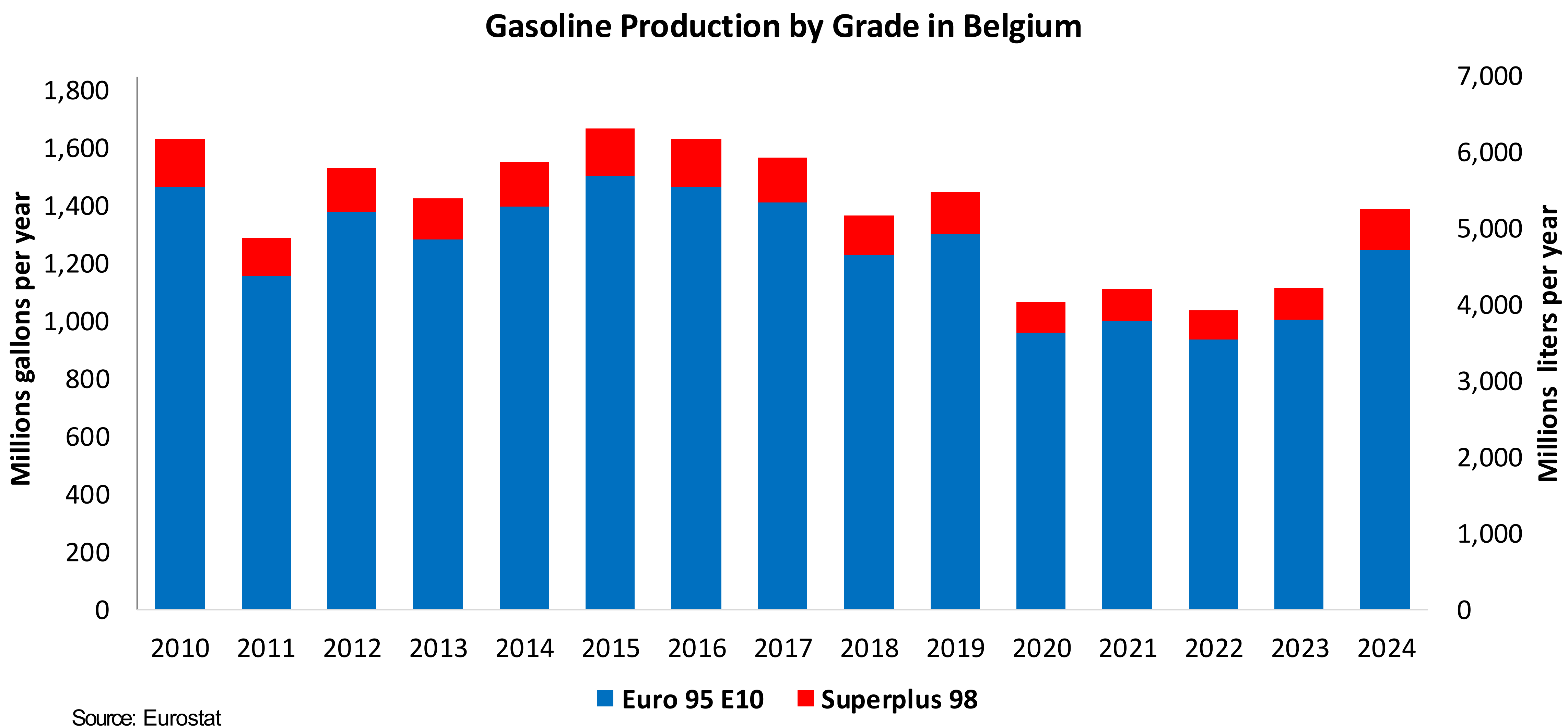
Source: Eurostat, European Commision

Gasoline Demand in Belgium



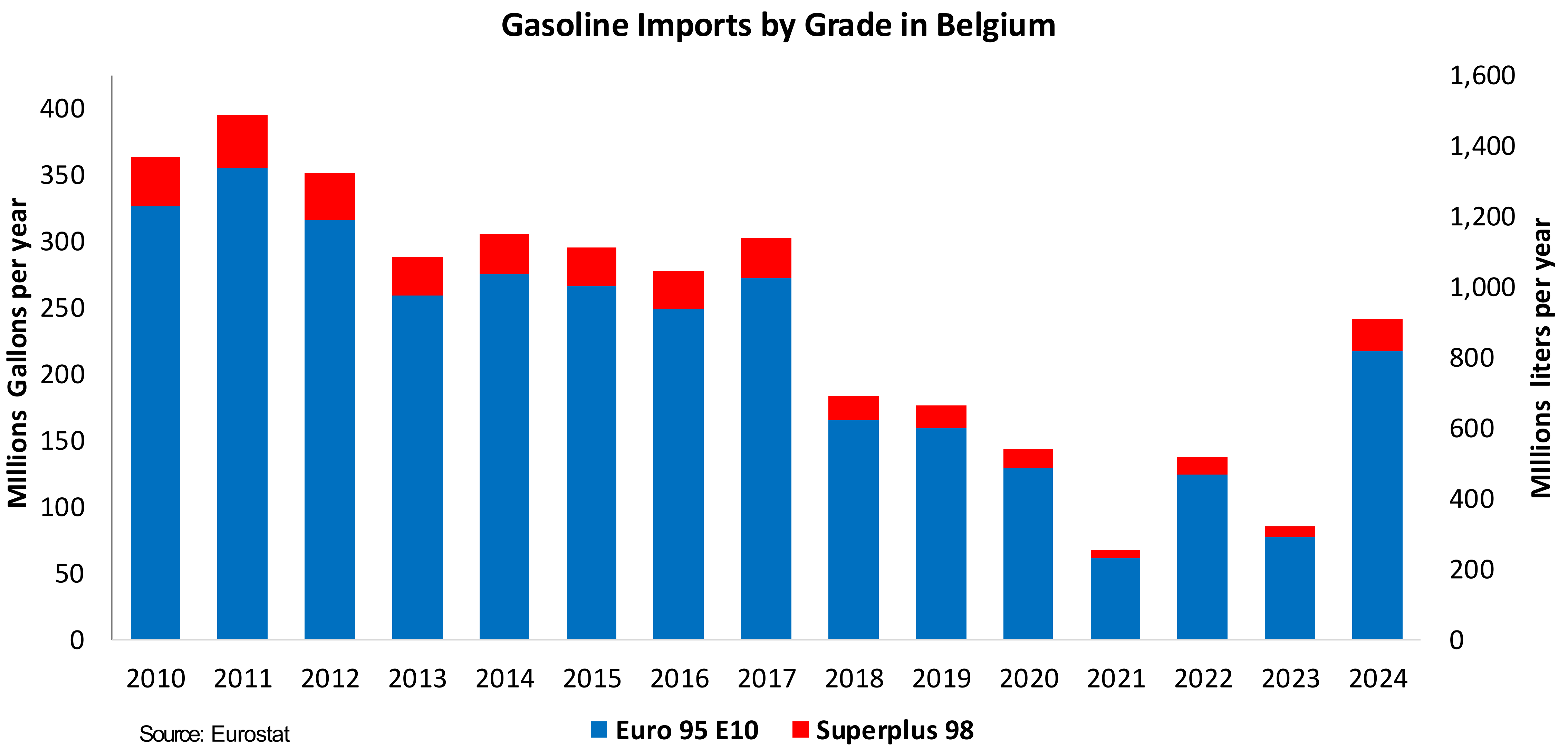
Source: Eurostat

Gasoline Production in Belgium



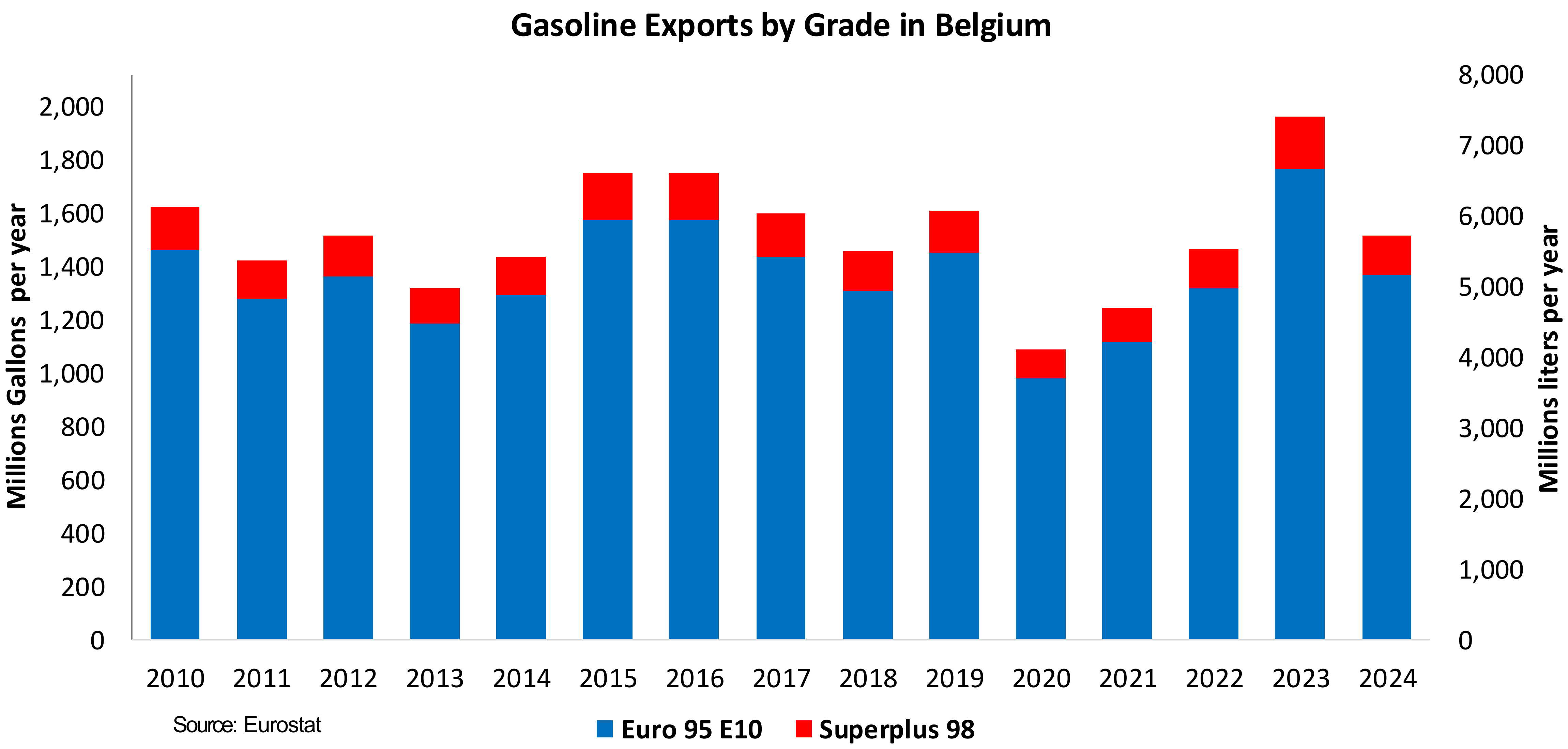
Source: Eurostat

Gasoline Imports in Belgium



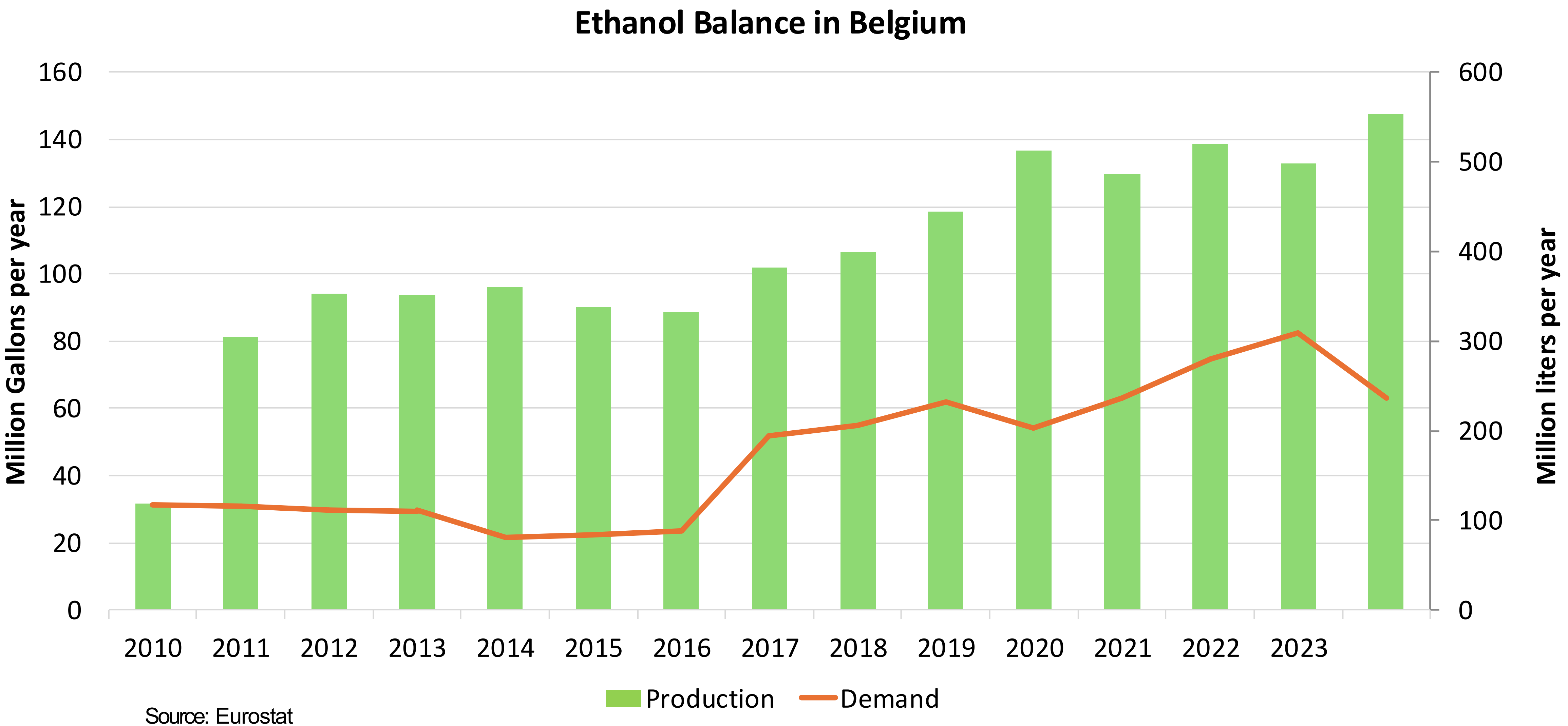
Source: Eurostat

Gasoline Exports in Belgium



Source: Eurostat

Ethanol Balance in Belgium



Source: Eurostat

Gasoline Quality in Belgium

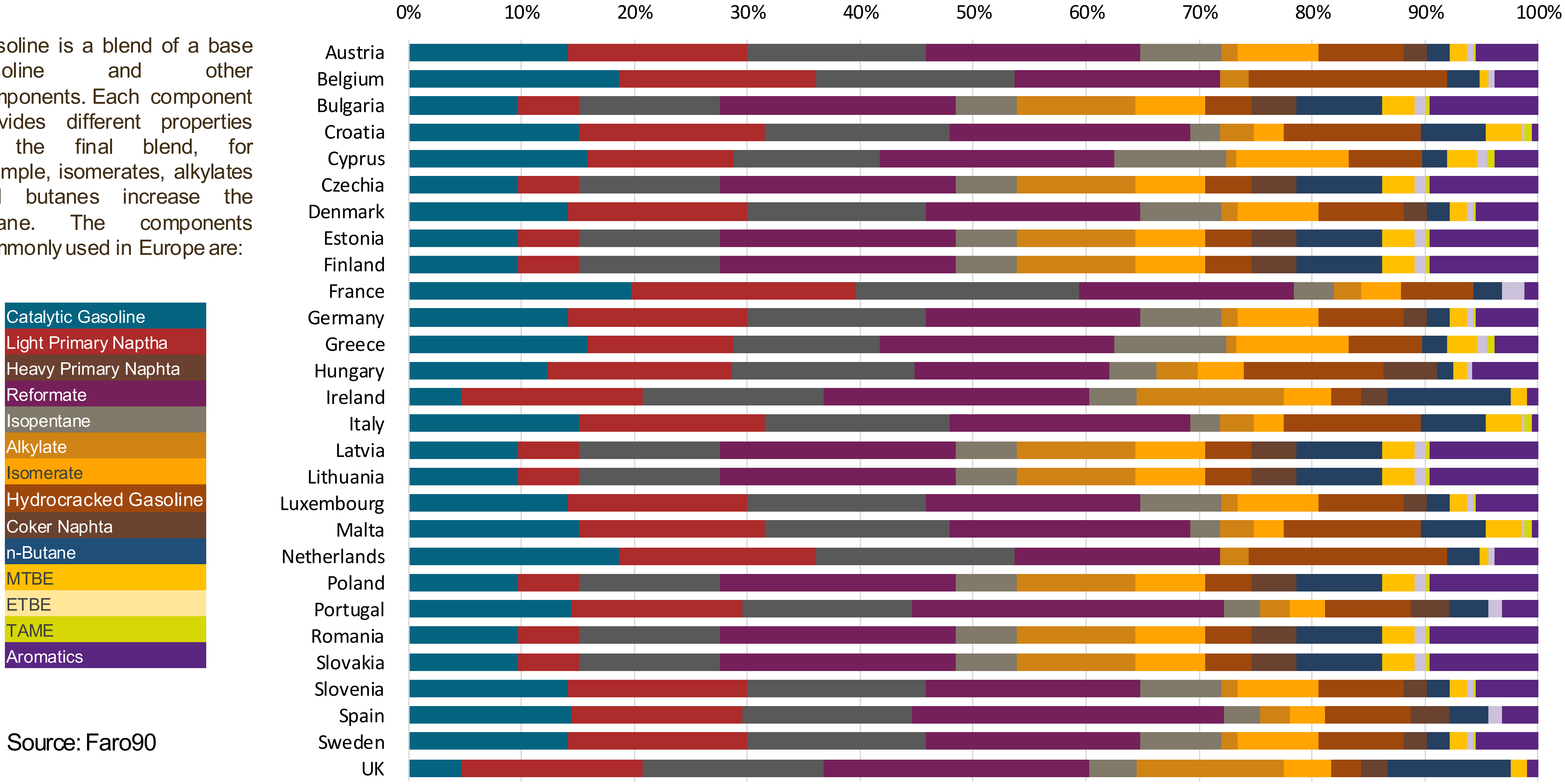
Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	10.5							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	5.7							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Gasoline Component Blending in Europe

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:



Source: Faro90

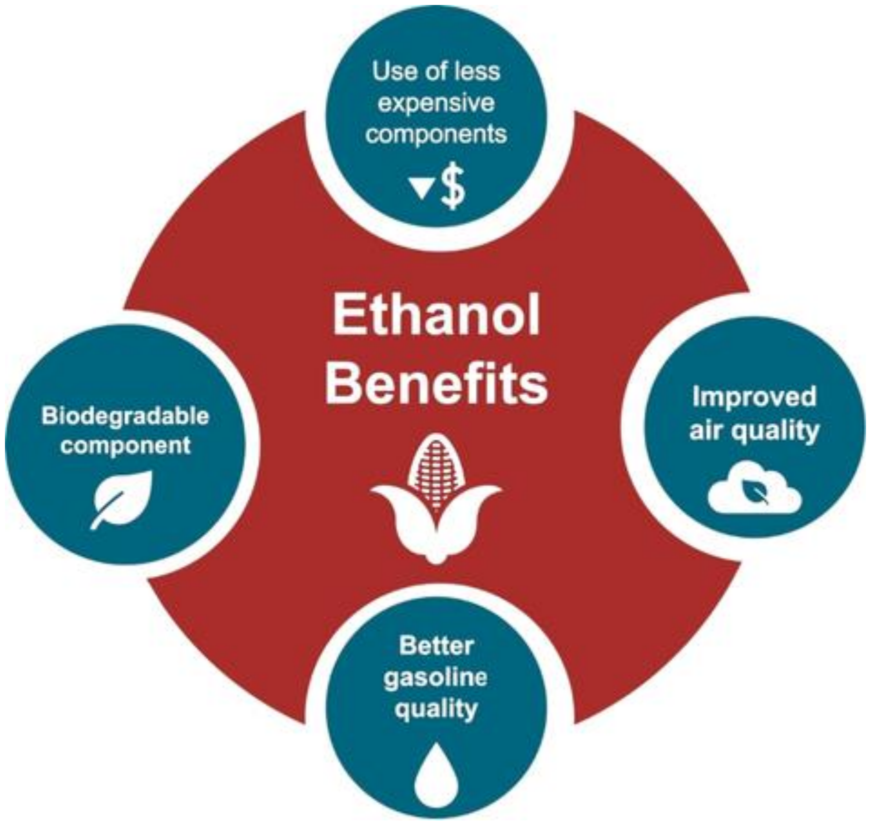
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

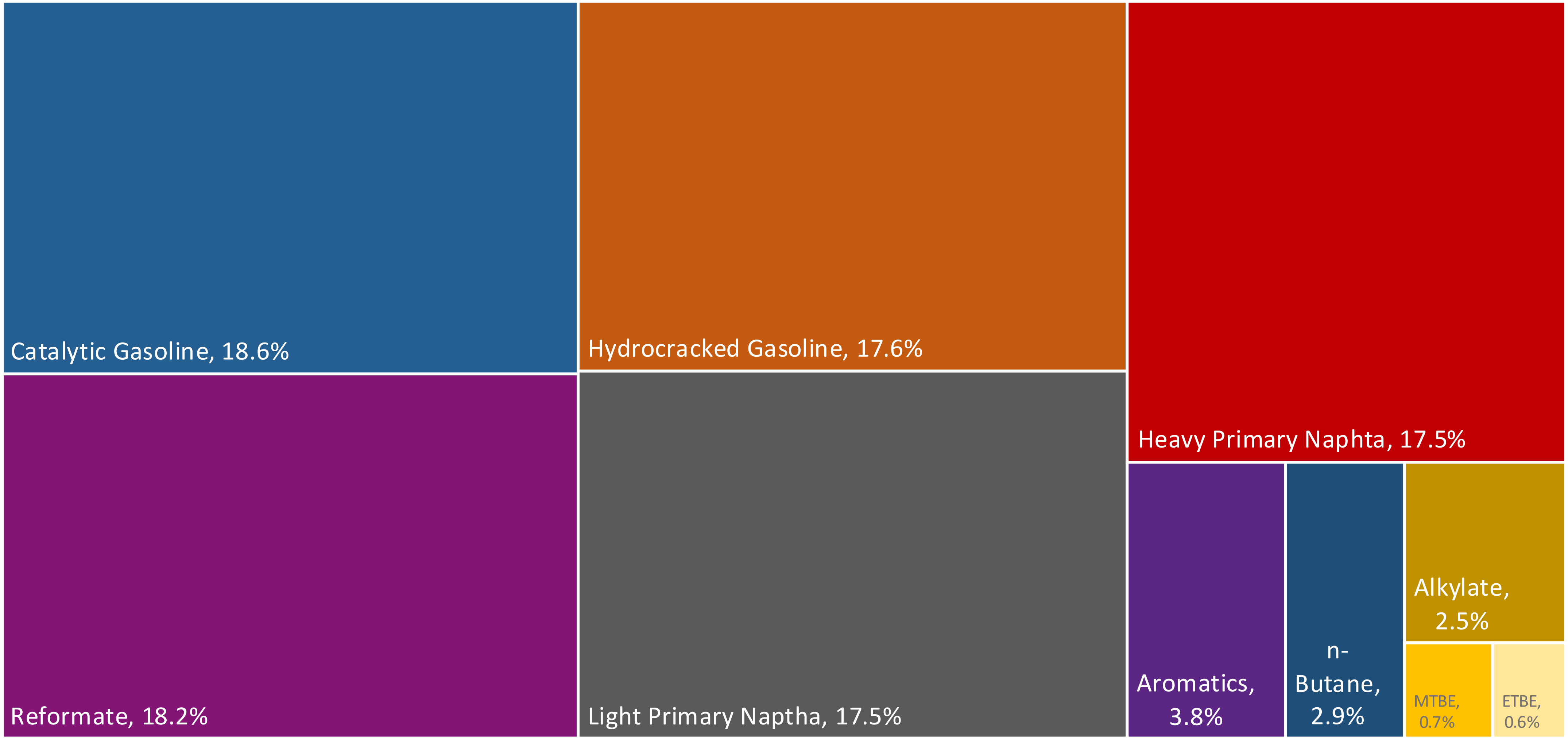
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

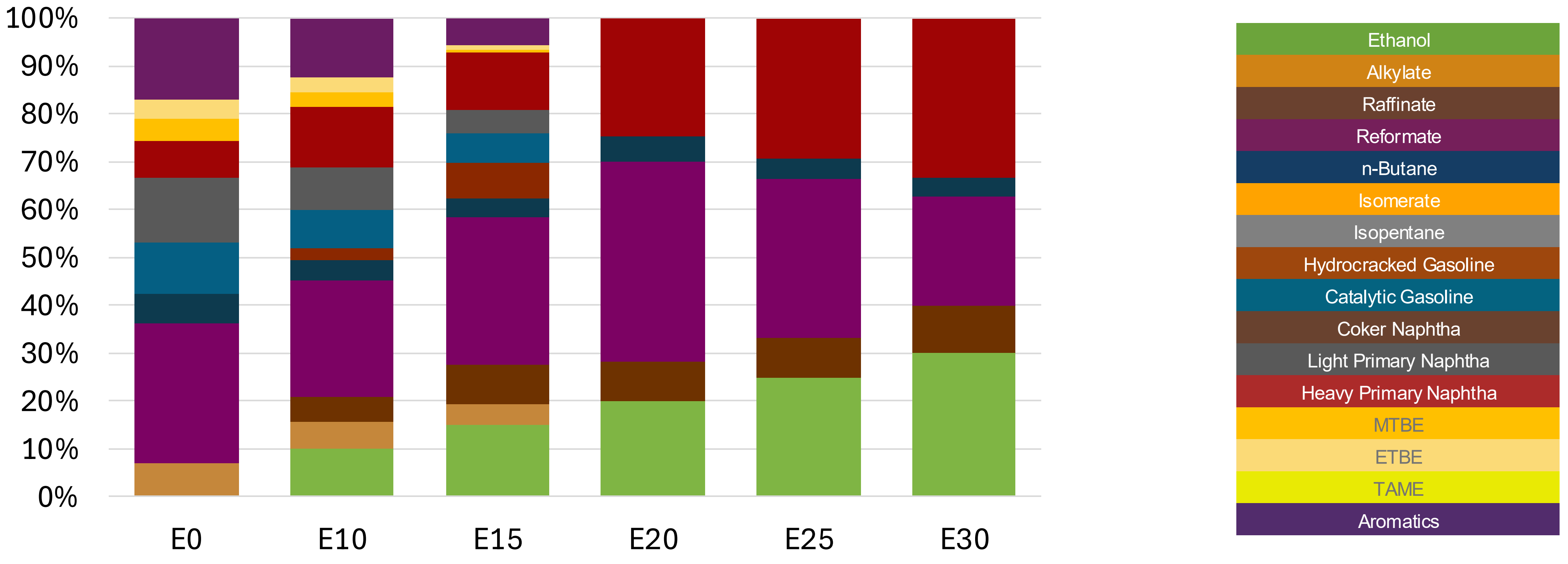


Belgium Available Blending Components



Source: Faro90

Belgium – Gasoline 95 – Constant Octane

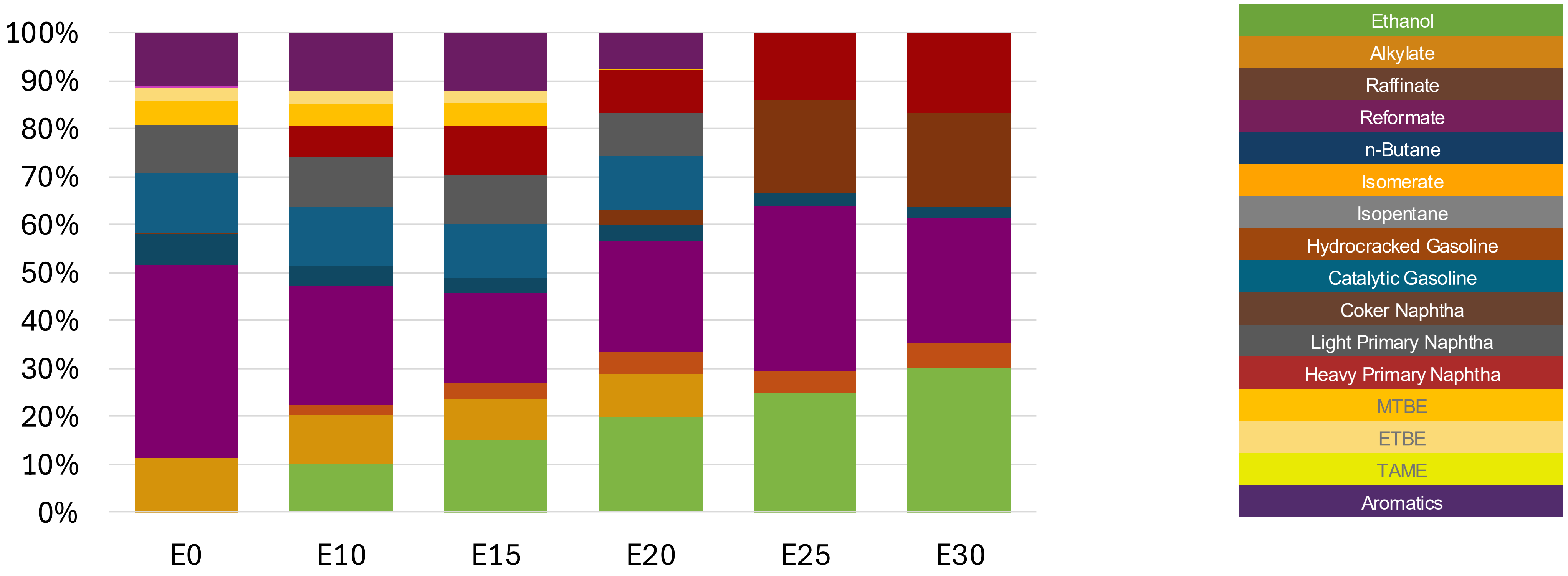


Octane (RON)	95.0	95.0	95.0	95.1	95.3	95.3
Price (US$/gal)	$ 2.281	$ 2.147	$ 2.072	$ 1.951	$ 1.875	$ 1.780

Average CIF 2024 Prices

Source: Faro90

Belgium - Gasoline 98 - Constant Octane

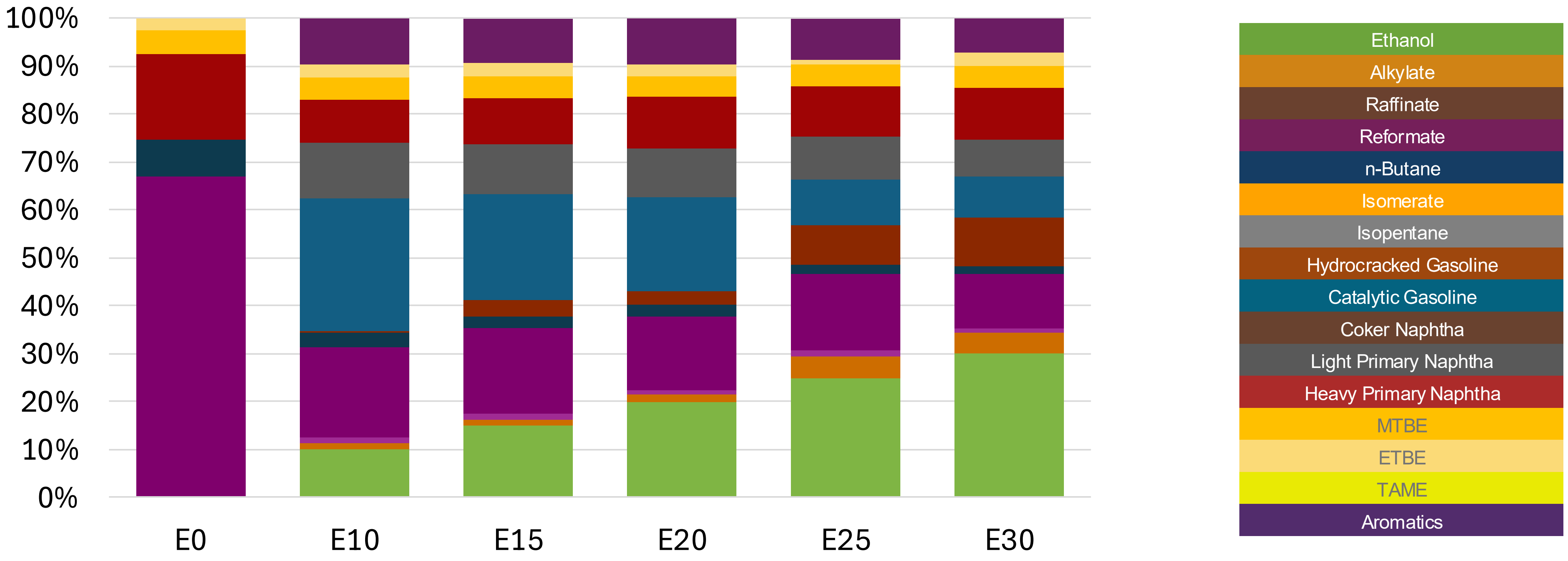


Octane (RON)	98.0	98.0	98.0	98.0	98.1	98.5
Price (US$/gal)	$ 2.409	$ 2.290	$ 2.216	$ 2.164	$ 2.057	$ 1.994

Average CIF 2024 Prices

Source: Faro90

Belgium – Gasoline 95 – Increased Octane

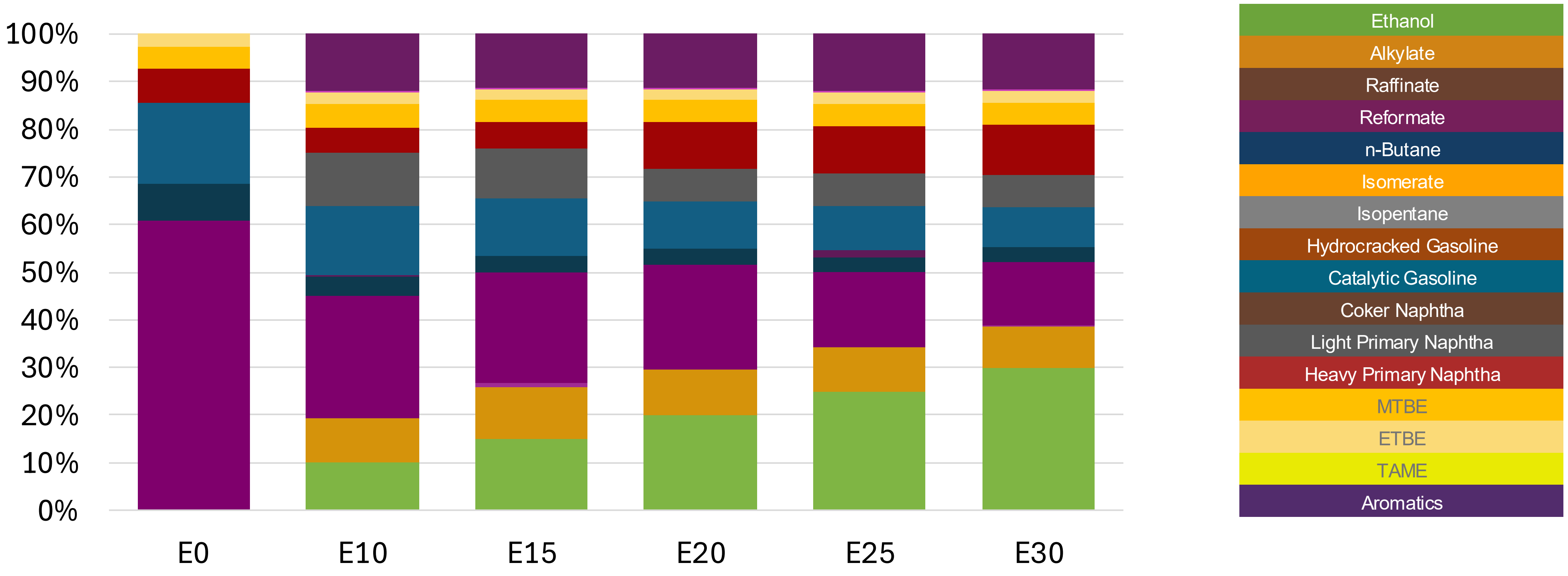


Octane (RON)	95.0	96.5	97.9	99.4	100.5	102.1
Price (US$/gal)	$ 2.193	$ 2.193	$ 2.193	$ 2.193	$ 2.193	$ 2.193

Average CIF 2024 Prices

Source: Faro90

Belgium – Gasoline 98 – Increased Octane



Octane (RON)	98.0	98.8	100.1	101.6	102.9	104.2
Price (US$/gal)	$ 2.329	$ 2.329	$ 2.329	$ 2.329	$ 2.329	$ 2.329

Average CIF 2024 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

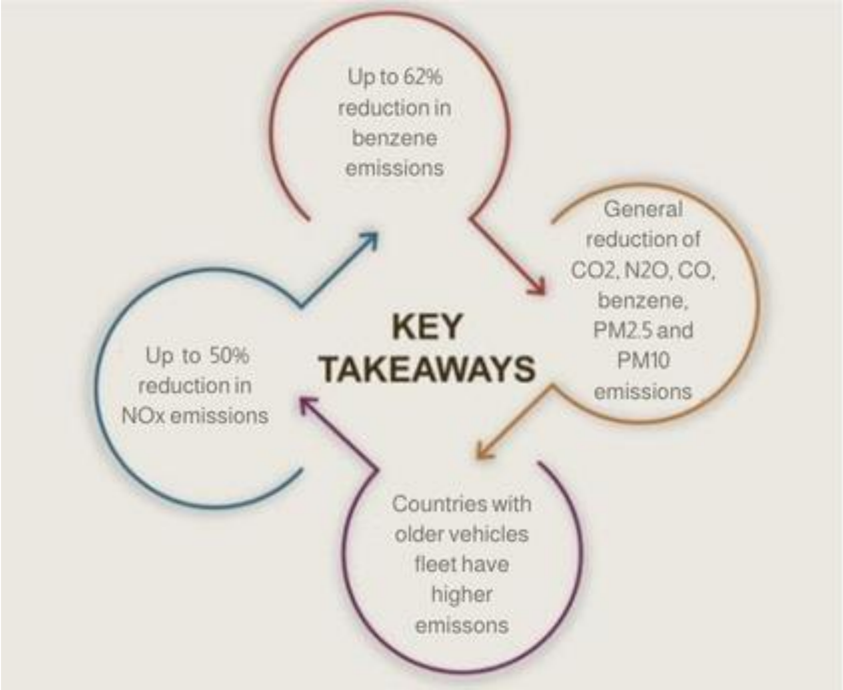
The model used in this analysis takes as a reference the [International Vehicle Emissions Model (IVE).]
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



Source: Bureau of transportation statistics.

Gasoline Vehicle Fleet - Belgium

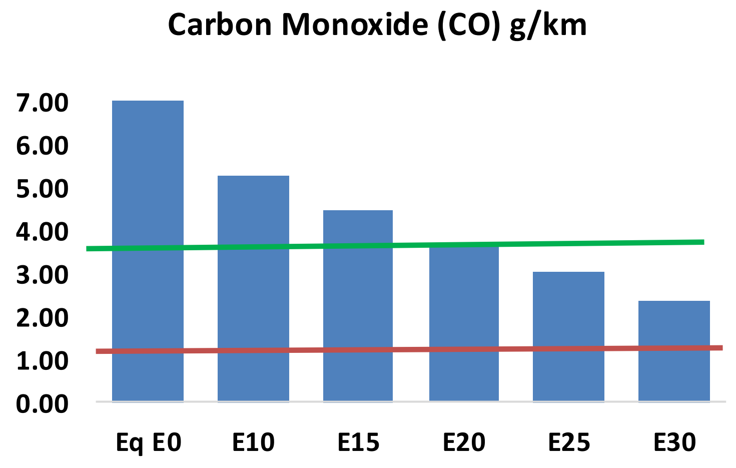
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	1,325
		Single FI	3-Ways		>8 years	7,652
		Multi FI	3-Ways		>8 years	157,086
			Euro III		>8 years	65,974
			Euro IV		>8 years	172,196
			Euro V		>8 years	3,547
			Euro 6		>8 years	126,866
					4-8 years	71,607
					<4 years	33,930
			Hybrid		All	142,544
		Electric			All	55,595
	Medium	Carburator	3-Ways	PCV Tank	>8 years	131
		Single FI	3-Ways		>8 years	172
		Multi FI	3-Ways		>8 years	14,786
			Euro III		>8 years	14,481
			Euro IV		>8 years	58,779
			Euro V		>8 years	383
			Euro 6		>8 years	42,289
					4-8 years	32,581
					<4 years	25,981
			Hybrid		All	85,156
		Electric			All	37,058
Sml Engines	Light	4-Cycle	None	PCV	>8 years	7,871
			Euro III		>8 years	13,387
			Euro IV		>8 years	51,663
			Euro V		>8 years	26,359
			Euro 6		>8 years	143,683
					4-8 years	36,132
					<4 years	598,985
			Hybrid		All	653,450
		Electric			All	265,894
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	16,574
		Fuel-Injection	3-Ways		>8 years	187,313
			Euro III		>8 years	105,052
			Euro IV		>8 years	121,096
			Euro V		>8 years	33,899
			Euro 6		>8 years	46,800
					4-8 years	22,064
					<4 years	2,867
			Hybrid		All	230
		Electric			All	271

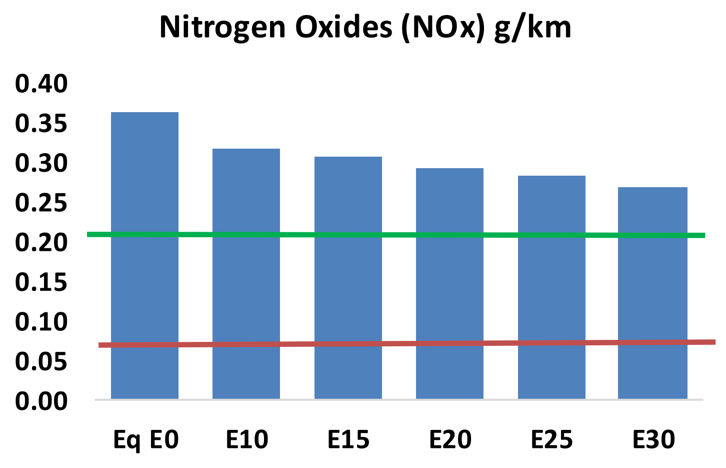
Vehicle Fleet: 3,483,706
Gasoline Vehicle Fleet: 2,367,805

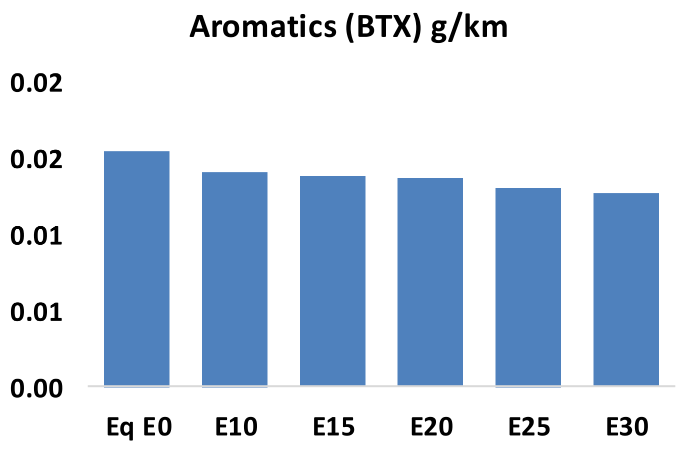
Source: Eurostat, ACEA

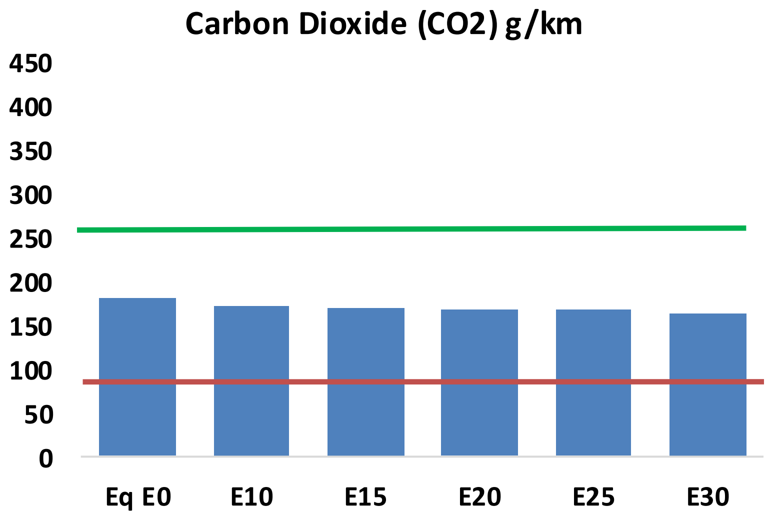
Belgium – Vehicular Emissions

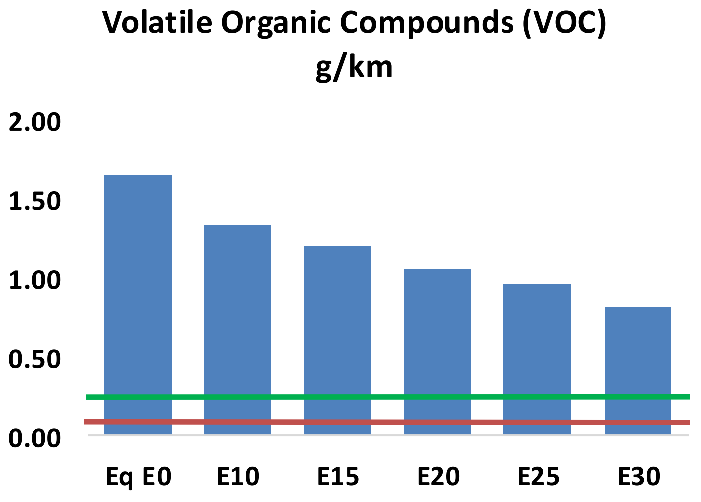
TIER III BIN 125 United States
Euro 6 Standard

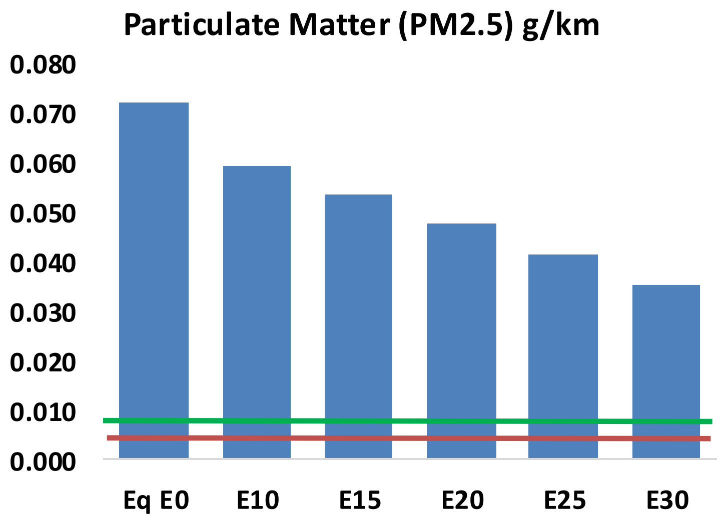












Source: Faro90

Belgium – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	424,560	371,491	318,422	270,948	221,338	182,637	143,912	-25%	-48%
CO2	10,967,384	10,685,366	10,419,015	10,209,538	10,105,881	10,047,600	9,862,394	-5%	-8%
VOC	99,964	90,398	80,831	72,624	64,196	57,665	49,129	-19%	-36%
NOx	21,975	20,290	19,175	18,628	17,723	17,025	16,196	-13%	-19%
SOx	124	115	105	97	90	81	73	-15%	-28%
NH3	4,858	4,810	4,762	4,785	4,839	4,877	4,908	-2%	0%
N2O	207	205	205	206	210	212	215	-1%	2%
CH4	19,986	19,986	19,986	20,285	20,725	21,105	21,465	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	403	375	347	324	302	285	261	-14%	-25%
Acetaldehyde	1,575	1,808	2,653	4,040	5,504	6,290	7,432	68%	249%
Formaldehyde	6,299	6,124	7,138	8,382	8,781	9,715	10,575	13%	39%
Benzene	934	897	850	837	830	793	768	-9%	-11%
PM2.5	669	608	547	494	442	385	327	-18%	-34%
PM10	3,696	3,360	3,024	2,728	2,439	2,129	1,804	-18%	-34%

Source: Faro90

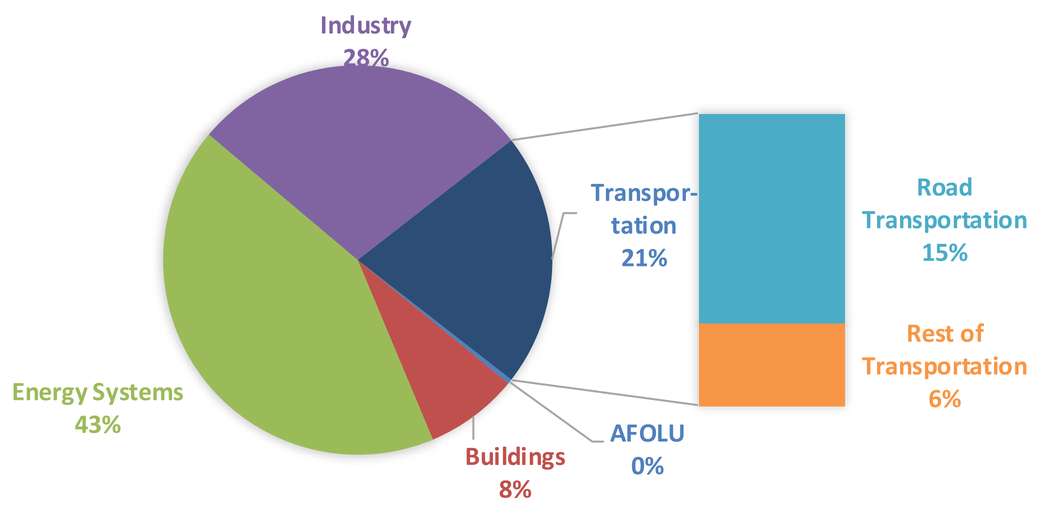
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

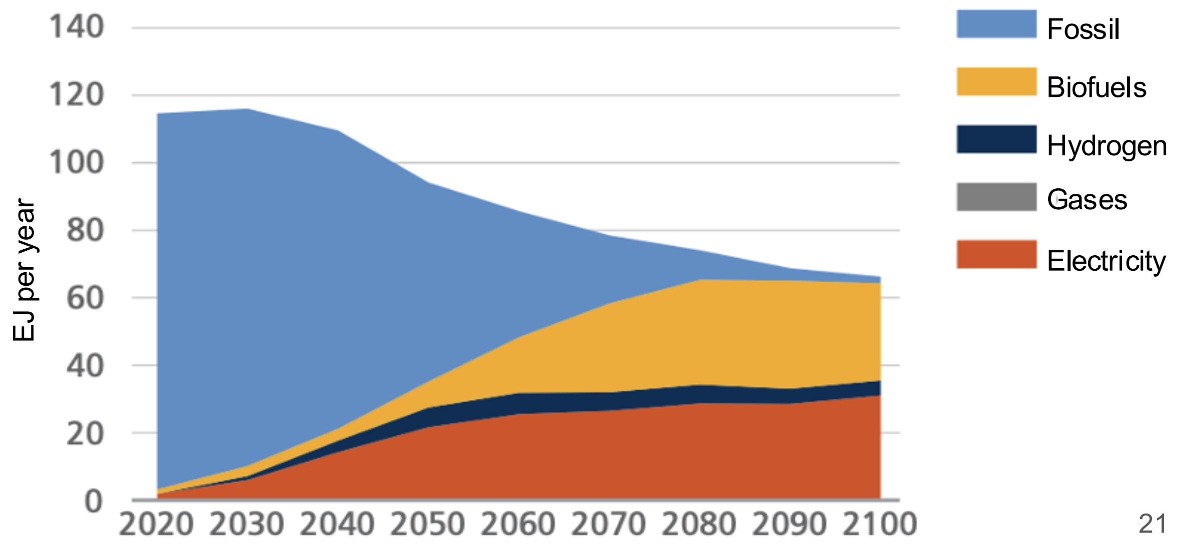
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

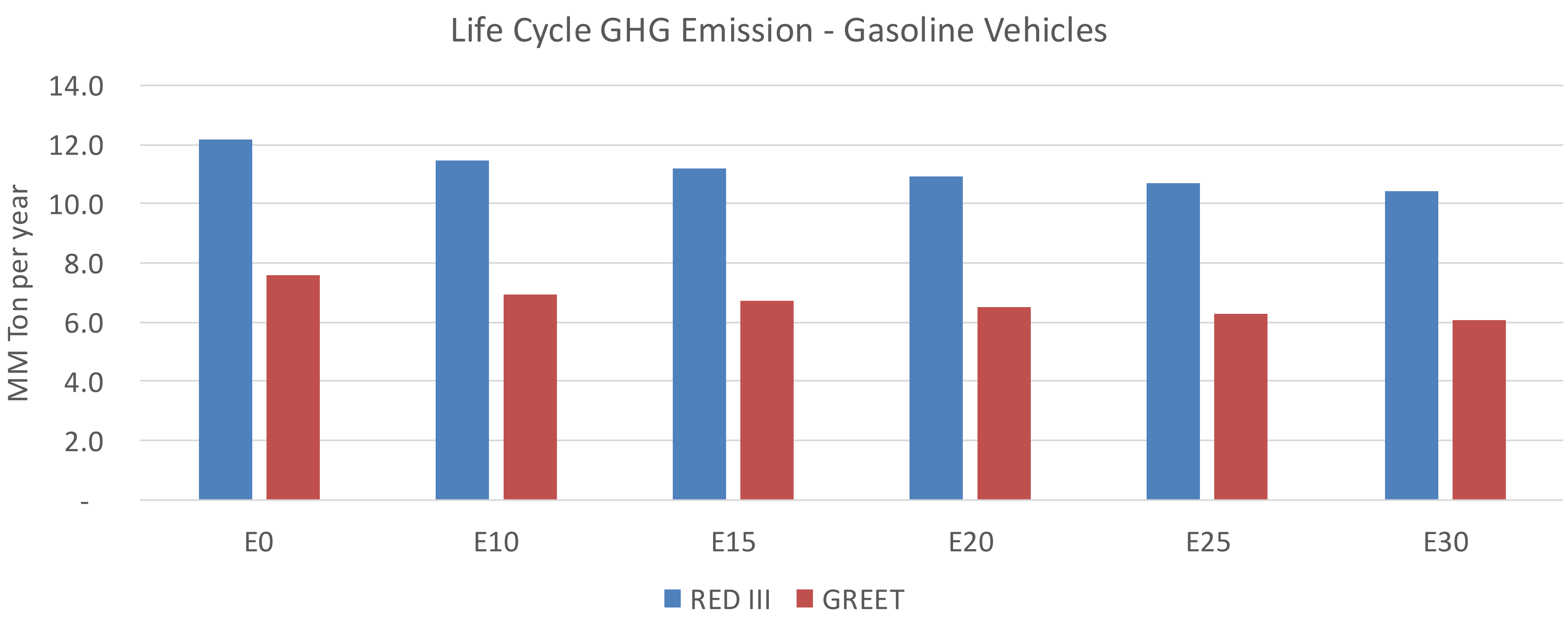
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Belgium – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	105.6
Target @ 2035 (MMT/year)	59.4
Delta	46.1

vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.8%	11.8%	14.7%	17.2%	20.5%
RED II/III	0.0%	6.2%	8.3%	10.4%	12.1%	14.4%

Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	1.5%	1.9%	2.4%	2.8%	3.4%
RED II/III	0.0%	1.6%	2.2%	2.7%	3.2%	3.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90